ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

COMPUTER ARCHITECTURE

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BSc Computer Science, 2nd year, semester 2 final examination. Section A: objective questions (10) · Section B: number systems and data representation, digital logic, the CPU and instruction execution, memory and caches, input/output and buses (90).

Written by Academic Central for practice, following second-year Computer Science syllabuses at South African universities (computer organisation and architecture). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Show ALL your working for conversions and calculations.
3. Non-programmable calculators may be used, but NOT for number-base conversions.
4. Number your answers exactly as the questions are numbered.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 What is the binary number 1011 in decimal?

A 9
B 11
C 13
D 15

1.2 What is the hexadecimal number 2F in decimal?

A 32
B 45
C 47
D 215

1.3 Which register holds the address of the next instruction to be fetched?

A Instruction register (IR)
B Program counter (PC)
C Accumulator
D Memory data register (MDR)

1.4 What is −1 in 8-bit two's complement?

A 10000001
B 11111110
C 11111111
D 00000001

1.5 Which logic gate gives an output of 1 ONLY when both inputs are 1?

A OR
B AND
C XOR
D NOR

1.6 Cache memory is …

A small, fast memory between the CPU and main memory.
B a large, slow store for backups.
C memory on the graphics card only.
D the part of the hard drive used for swapping.

1.7 How many memory locations can a 16-bit address select?

A 16
B 256
C 65 536
D 4 294 967 296

1.8 In the instruction cycle, which step comes first?

A Decode
B Execute
C Fetch
D Store

1.9 In the von Neumann architecture …

A instructions and data are stored in the same memory.
B instructions and data have separate memories.
C there is no main memory.
D programs run only from ROM.

1.10 Which of these loses its contents when the power is switched off?

A ROM
B RAM
C A hard disk
D Flash memory

[10]

SECTION B

QUESTION 2: NUMBER SYSTEMS AND DATA REPRESENTATION

2.1 Convert the decimal number 156 to binary and to hexadecimal. (4)

2.2 Convert the binary number 1101.101 to decimal. (2)

2.3 Represent −45 in 8-bit two's complement. (3)

2.4 Add the 8-bit two's complement numbers 01100100 and 01010000. Give the 8-bit result and say whether overflow occurred. Explain. (3)

2.5 Represent 6.75 in IEEE 754 single-precision floating point. Show the sign bit, the 8-bit exponent and the 23-bit mantissa. (5)

2.6 Write your 32-bit answer to 2.5 in hexadecimal. (2)

2.7 Why do programmers often write binary values in hexadecimal? (1)

[20]

QUESTION 3: DIGITAL LOGIC

3.1 Draw the truth table for F = A·B + A′·C (A′ means NOT A). (4)

3.2 Use Boolean algebra to simplify F = A·B + A·B′ + A′·B. Name the law used in each step. (4)

3.3 Use a Karnaugh map to simplify F(A, B, C) = Σm(0, 2, 4, 5, 6). (5)

3.4 Draw a logic circuit for your answer to 3.3, using NOT, AND and OR gates. (3)

3.5 Give the truth table of a half adder, and the Boolean expressions for its Sum and Carry outputs. (4)

[20]

QUESTION 4: THE CPU AND INSTRUCTION EXECUTION

4.1 Name FOUR registers in the CPU and give the purpose of each. (4)

4.2 Describe the fetch–decode–execute cycle using register-transfer steps (for example MAR ← PC). (5)

4.3 A simple accumulator machine runs the program below. Memory location 20 holds 7, 21 holds 5 and 23 holds 4. Trace the program: give the value of the accumulator (ACC) after each instruction, and the final values in locations 22 and 24. (4)

Address	Instruction	Meaning
0	LOAD 20	ACC ← M[20]
1	ADD 21	ACC ← ACC + M[21]
2	STORE 22	M[22] ← ACC
3	SUB 23	ACC ← ACC − M[23]
4	ADD 22	ACC ← ACC + M[22]
5	STORE 24	M[24] ← ACC
6	HALT	stop

4.4 Give FOUR differences between RISC and CISC processors. (4)

4.5 Explain what pipelining is, and describe ONE type of pipeline hazard. (3)

[20]

QUESTION 5: MEMORY AND CACHES

5.1 Arrange these from fastest to slowest: main memory (RAM), registers, hard disk or SSD, cache. Explain why computers use such a memory hierarchy. (4)

5.2 Explain temporal locality and spatial locality. Give an example of each in a program. (4)

5.3 A computer uses 32-bit byte addresses and a direct-mapped cache with 64 lines of 16 bytes each. How many bits of the address are used for the offset, the index (line) and the tag? Show your working. (6)

5.4 A program makes 1 000 memory accesses, and 950 of them are cache hits. The cache access time is 2 ns and main memory takes 50 ns more on a miss. Calculate the hit ratio and the average (effective) memory access time. (4)

5.5 What is virtual memory, and why is it used? (2)

[20]

QUESTION 6: INPUT/OUTPUT AND BUSES

6.1 Name the THREE buses that make up the system bus, and say what each one carries. (3)

6.2 Explain the difference between polling and interrupts for input/output. (3)

6.3 What is direct memory access (DMA), and why does it improve performance? (2)

6.4 A computer has a 32-bit address bus and byte-addressable memory. What is the largest amount of memory it can address? Show your working. (2)

[10]

TOTAL: 100

END OF QUESTION PAPER